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CO5151 – Advanced Agentic Artificial Intelligence

D2 Interim Report

LuatCheck: A Local, Source-Grounded Assistant over Vietnamese Regulations

Application Track – Group 8

[PENDING: Draft of 24 October 2026. Result tables are generated from the two frozen v3 runs; the human audit of the judge, the post-hoc revision r2 and the team's inputs are still pending.]

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Repository: https://github.com/tuandung222/LuatCheck-D2

Reported revision: **[PENDING: Insert the full final commit hash after freezing the source.]**

Access: **[PENDING: Confirm the instructor can read the repository.]**

Status: At risk (provisional). The benchmark design is frozen, the pipeline is in place and the v3 main runs exist (Section 5); strict success is low for every variant, and the human audit of the judge, the post-hoc revision r2 and the demo rehearsal are outstanding.

HO CHI MINH CITY, OCTOBER 2026

Summary. LuatCheck is a local agent that answers only from a document collection, cites the sections it used, checks those citations in code, and refuses or asks a clarifying question when the sources do not answer. One bounded agent loop drives a 2B ternary controller that fits a 16 GiB Mac.

Implemented: the loop, a read-only tool layer with gated writes, lexical citation checking (L1) with bounded repair, an optional cross-encoder reranker, two ingested corpora (academic regulations of the Ho Chi Minh City University of Technology, HCMUT, and Vietnamese labour-law texts) and a one-command, hash-locked benchmark pipeline (Sections 2–4).

Evaluation (frozen design, v3 runs collected and judged): two 50-task benchmarks × three variants (OFF: no verification; L1: lexical verification with repair; L1+rerank: L1 plus reranker) × three seeds, plus two diagnostic baselines (closed-book and oracle, one seed, not system variants), scored by a large language model (LLM) judge whose reliability is to be checked against human labels (Section 5).

Main result (Bonsai 2B, LLM-judge lower bounds): strict success is 12.0–18.7% for every variant on both benchmarks; L1 and the reranker change it by no detectable amount, L1 lowers the judge-free behaviour match by 20.7 and 22.7 pp, and the oracle with gold evidence reaches only 28.9% and 44.4%.

Pending, in red throughout: the judge audit, the post-hoc revision r2 (not yet run), member contribution evidence, the commit hash and the demo rehearsal. Self-assessment: **at risk** (Section 7).

1 Changes from the Proposal

Scope and users. The proposal (D1, Application Track) targeted employees and small-business HR staff asking labour-law questions. For D2 we added a second, non-legal corpus: the academic-affairs regulations of HCMUT, used by students and advisors. It tests whether the code is domain-agnostic, since domain wording, tool names and ID rules live in `config/domains/` as data. We claim only what is evaluated: ten HCMUT regulations and ten labour-law documents (the 2019 Labour Code and nine laws, decrees and circulars of 2020–2025), snapshotted on 22 October 2026 with later amendments unchecked. LuatCheck is not legal advice and has not been tried by real users.

Scenarios. (1) A student looks up a rule (e.g. make-up class registration) and opens the cited section. (2) An employee or HR officer asks a labour-law question and checks the cited article. (3) A careless or malicious user pastes a document with injected instructions, or asks how to evade a rule; the system must hold its tool boundary and clarify or refuse. These are intended uses, not user-study results.

Commitments and their state. Table 1 compares the D1 commitments due at the D2 checkpoint (week 7) with their current state.

Proposal commitment (D1)	State	Delta and reason
Budgeted loop, schema output, L1 citation check	Done	Runs end to end (Section 2); budgets and failure semantics unchanged.
Qwen3-4B/8B controllers; 32B comparator	Dropped	The controller is Ternary Bonsai 2 2B (GGUF). It fits a 16 GiB Mac and one GPU slot, so the same weights serve the benchmark and the demo. A 20-trial diagnostic on the earlier task set ran about 2× faster than the MLX build (`gguf2b-bonsai-diag-v1`).
BM25 + Qdrant dense retrieval; 200 held-out VLQA pairs	Dropped	Replaced by BM25 + TF-IDF fused by reciprocal rank fusion (RRF). The persisted vector index exists but is not used in D2 runs. The VLQA record mapping could not be established, so retrieval is measured as source-ID metrics on the new benchmarks.
≥10 end-to-end tasks; 3 configurations × 3 seeds	Done	Two frozen 50-task benchmarks, variants OFF / L1 / L1+rerank, seeds 42, 123 and 2026 (900 system trials), plus 190 diagnostic-baseline trials; collected and judged as two runs of 545 trials (Section 5).
Reranker (optional in D1)	In progress	Implemented and wired as variant L1+rerank; parity of its two runtimes pending (Section 4).
L2 entailment with natural language inference (NLI), optional	In progress	Implemented as optional components; excluded from all D2 runs.
Guarded `send_summary`, UI, audit log	Done	Changed: the model can only propose; a trusted UI approves an exact payload; reversible local outbox, no sender.
Verifier audit (30 pairs)	Not started	Changed: replaced by a human audit of the LLM judge (about 30 items per benchmark), scheduled for Sat 24 Oct: **[PENDING]**.
Threat model v1	Done	Updated in Section 6.
First injection results	Not started	Not required in D2; planned for W47.
HR walkthrough (W5)	Not started	Reported by the group on 22 October. **[PENDING: New date and owner.]**

Table 1: D1 commitments due at the D2 checkpoint and their state (done, in progress, not started, dropped).

Feedback and earlier pilot. The group received no feedback on the D1 pitch (22 October), so no change is attributed to feedback. An earlier pilot (Qwen2.5-3B on Ollama, ten-task `bll2019_v1`) is superseded; none of its numbers appears here.

Models and framework. Table 2 lists the components behind the reported numbers. We use no agent framework. The loop, action validation, tool registry, verifier, ledger, memory and harness are our own code (AI-assisted

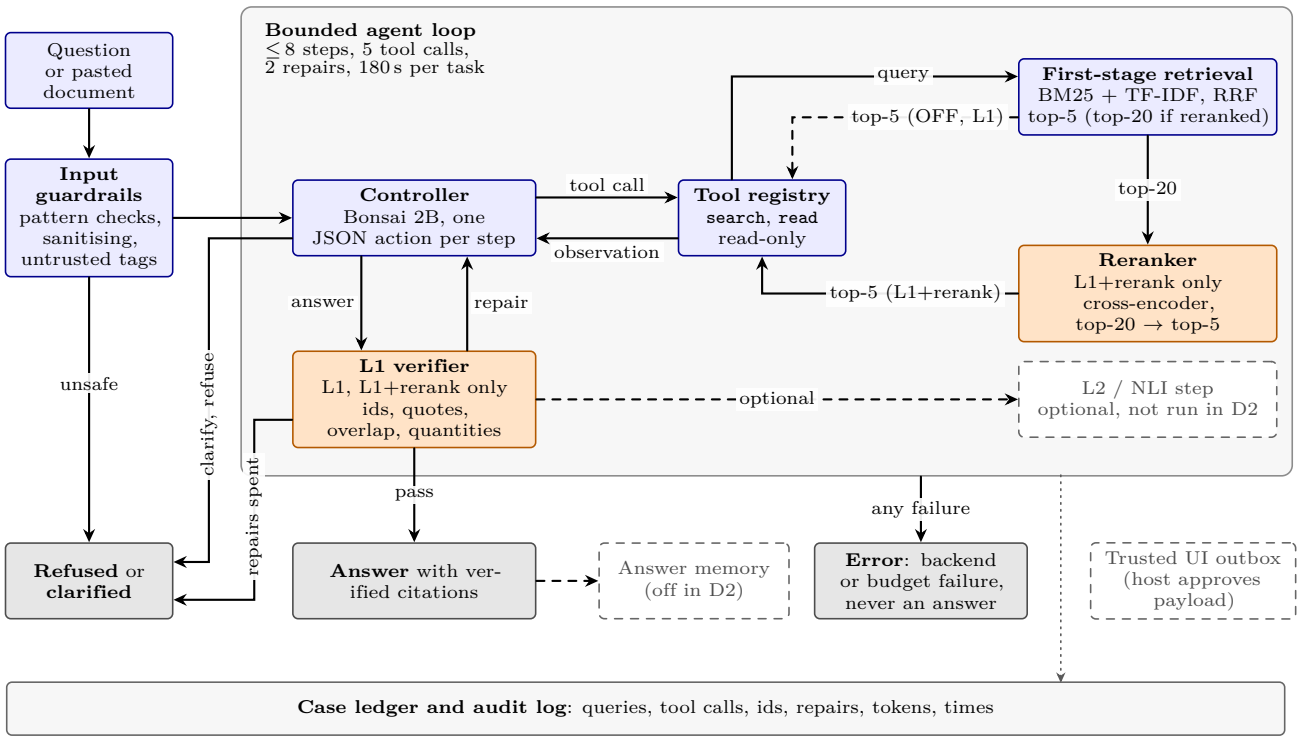


Figure 1: Implemented architecture and control flow. Blue boxes run in every D2 variant; orange boxes run only in the variants they name (OFF skips the verifier; without the reranker, retrieval returns the top five directly); dashed boxes are implemented but outside every D2 run; grey boxes are the terminal states of a run. The dotted arrow marks that every step is written to the ledger.

coding is disclosed in Section 8), built on `rank-bm25`, scikit-learn, `pypdf`, the Model Context Protocol (MCP) SDK, FastAPI/React and Streamlit. Controller sampling is fixed: temperature 0.7, top-p 0.8, top-k 20, presence

Component	Choice	Runtime and role
Controller	Ternary Bonsai 2 2B, GGUF, PQ2_0 quantisation (`prism-ml/Ternary-Bonsai-2-2B-gguf`; file SHA-256 `d5f01a4e`…)	PrismML fork of `llama.cpp`, commit `73bfe6c`; Metal on the Mac, CUDA on server 82 (build `b10787-73bfe6cc4`; the PQ2_0 file ran on CUDA in both v3 runs). Used for every reported number and for the live demo.
Reranker	`AITeamVN/Vietnamese_Reranker`, cross-encoder, revision `f5369762` (weights SHA-256 `1cecccab`…)	Variant L1+rerank only: torch fp16 on server 82, a GGUF conversion on the Mac.
First-stage retrieval	BM25 + TF-IDF fused by RRF; no neural embeddings	In memory; all variants.
Judge	DeepSeek V4.1 Flash	Through an API gateway; offline evaluation only.

Table 2: Models and software behind the reported numbers. Server 82 is the shared GPU server (one H100 used).

penalty 1.5, decoding constrained to the action JSON schema, thinking off, at most 1,024 output tokens. The live demo runs the same model file, `llama.cpp` commit, sampler, prompts, budgets and corpora, one query at a time through the web UI. It illustrates the tables; it is not evidence for them.

2 Implemented Architecture (R1)

LuatCheck is a single agent running a ReAct-style loop [8] over retrieval-augmented generation [6] (Figure 1). At each step the controller emits one JSON action: `search`, `read`, `answer`, `clarify` or `refuse`. The Python tool registry executes searches and reads and returns the result as an observation. An `answer` passes through the verifier; if the check fails, the verifier's feedback goes back to the controller for at most two repairs, which is external feedback rather than self-critique [5]. Input guardrails refuse unsafe input before any model call. The loop is bounded to 8 steps, 5 tool calls, 2 repairs and 180 s per task (60 s per model request). A backend or model-format failure ends the run as `error`; it is never turned into an answer or a refusal.

There is one controller and no multi-agent coordination, which keeps permissions and budgets easy to inspect. The MCP server exposes the same tools but the loop does not use it. The three variants test whether verification and reranking help (Section 5).

What runs end to end. Every D2 run uses input guardrails, the controller loop, `search` and `read_document`, first-stage retrieval, the ledger and the audit log. The L1 verifier with repair runs in L1 and L1+rerank, and the reranker only in L1+rerank. Four components are implemented but outside all D2 runs: the L2/NLI step, the

persisted hybrid index with dense retrieval, answer memory and the trusted-UI outbox. An external sender and code execution do not exist.

Real task trace. A trial from the main run will be rendered from its ledger (`trials.jsonl`) as Table 3. **[PENDING: Select trial id and run id; fill the table from the recorded steps only.]**

Step	Action	Tool call / result ids	Verifier verdict	Tokens in/out	Time (s)
1	[PENDING]	[PENDING]	[PENDING]	[PENDING]	[PENDING]
2	[PENDING]	[PENDING]	[PENDING]	[PENDING]	[PENDING]
...	[PENDING]	[PENDING]	[PENDING]	[PENDING]	[PENDING]
Total				[PENDING]	[PENDING]

Table 3: Trace of one completed main-run trial (task, variant, seed, run id: [PENDING]). Placeholder rows: nothing here is a recorded step.

3 Tools and Permissions (R2)

Table 4 lists every tool and model-facing interface with its state and permission level. The controller's action space (`ActionType`) contains no write action, so it has no direct write path. A stdio MCP check confirmed discovery, search, lookup, blocked unverified writes and proposal-only summaries (`docs/reviews/native-integrity-revision-2026-10-22`). It tests the tool contract, not a model task. The demo serves one user in one process; multi-user isolation is not certified. **[PENDING: Attach application demo evidence from the rehearsal.]**

Interface	State	Permission level	Enforced boundary
`search`, `read_document` (labour profile: `search_law`, `get_article`)	Implemented	Read-only	Return local corpus text only.
Reranker (inside `search`)	Implemented	Read-only (model component)	Scores candidates; a failure is an execution error (`ModelComponentError`), never a first-stage fallback.
Answer-memory write	Implemented; off in D2	Reversible write	SQLite write only after the verification gate.
MCP `save_case`	Implemented; blocked	Reversible write	A model caller cannot self-attest verification; the call returns blocked.
`send_summary`	Proposal only	Irreversible write	The model-facing tool can only create a proposal; it cannot approve or execute.
Trusted-UI outbox (`dispatch_external_action`)	Implemented	Irreversible write (local queue only)	Needs host approval and a token bound to the payload hash; a caller flag is insufficient; a queue receipt is not a delivery.
External sender; code execution	Not implemented	Irreversible write; code execution	No e-mail or webhook transport and no code-running tool exists.

Table 4: Tool inventory (`src/tools/permissions.py`, `domain_tools.py`). Permission levels are those classified in code, and every non-read-only interface is blocked or gated there, not just in the prompt.

4 Memory and Knowledge Sources (R3)

Short-term context. Each run keeps one message list: the domain system prompt, the question (and any pasted document) wrapped as untrusted input, then alternating action and tagged observation turns. Observations are compacted. `search` returns at most the two clauses that best match the query, plus a truncation flag; a repeated query or re-read returns a short notice; budget notices tell the model to answer or refuse. The benchmark uses an 8,192-token context, and the largest controller input in the smoke runs was 3,944 tokens. The ledger records queries, retrieved text and ids, budgets and per-call telemetry.

Episodic memory (implemented, not exercised in D2). Verified answers can be stored in SQLite after the verification gate. Reuse requires a matching index version, native-origin metadata and a question similarity of at least 0.85 (TF-IDF cosine and Jaccard), and the reused answer is re-verified. Memory is off in every D2 run, so trials stay independent. Reuse is also off in the demo, because matching sees only the question and not the facts that decide a case. The filter for personally identifiable information (PII) covers a fixed set of patterns.

Knowledge sources. `src/ingestion` built both corpora from public PDFs: `pypdf` text extraction, Unicode normalisation and segmentation at section headings. Each record carries its source URL, file SHA-256 and page numbers (manifests in `docs/materials/*-sources`), and IDs have the form `<DOC>_<section>`. Table 5 gives size and known limits.

Retrieval and reranking. Search runs in memory: BM25 over title and content and TF-IDF over word 1–3-grams, fused by RRF ($k = 60$). Variants OFF and L1 return the top five first-stage results. L1+rerank takes

Corpus	Documents	Records	Known limits
`hcmut_academic_v1`	10	143	Several documents are amendment-only; the image-only base regulations are absent; one record has about 19,600 characters; printed decision numbers come from a web listing.
`labour_vi_v1`	10	458	Decree 145/2020 (over 40 pages) and unnumbered annexes are absent; texts as promulgated, later amendments unchecked.

Table 5: Corpora used in D2 (`data/corpus/`, snapshot of 22 October 2026). Neither has been certified by a domain expert: **[PENDING: university or legal reviewer, if any]**.

the top 20 candidates, rescores them with the cross-encoder and returns the best five. Long records would lose late evidence if truncated, so a record longer than one window is split into overlapping windows (about 1,500 characters, 200 overlap). Each (query, window) pair is scored and the record keeps the maximum (MaxP).

The reranker's input policy and weights are identical on both machines, but the runtimes differ: torch fp16 on CUDA on server 82, and a GGUF conversion on the `llama.cpp` server on the Mac (quantisation: **[PENDING: confirm after the parity check]**). Scores are therefore not bit-identical: **[PENDING: rank and score parity between the two runtimes, from the parity probe]**. Accuracy comes from the GPU run and latency from the Mac run; the two L1+rerank rows are not pooled. The persisted hybrid index and dense retrieval (`src/indexing`) are outside D2.

5 Evaluation Set and Initial Results (R5)

Benchmarks and provenance. `hcmut_v1` and `labour_vi_v1` each hold 50 Vietnamese tasks: 40 single-record, 5 two-record and 5 out-of-scope tasks that expect a refusal. Each task has gold record IDs, a short reference answer and verbatim evidence quotes, and both files are frozen with hashes (`benchmark.lock.json`). **No human wrote or reviewed any task.** Claude Haiku drafted them; two independent Claude Sonnet verifiers checked every task against the cited records; a Sonnet adjudicator fixed or replaced flagged items; mechanical checks confirmed that quotes are verbatim and IDs exist; and Gemma-4-31B answers given the gold context served as a sanity pointer (LLM estimates, not labels). The items are AI-written questions over real documents, not real user questions; real use (long, vague, pasted contracts) will differ. **[PENDING: Member spot-check against the source PDFs, if done.]**

A later AI audit (`docs/reviews/benchmark_quality_2026-10-23.md`) found no wrong gold answer and no refusal item answerable from the corpus. It did flag three limits:

- **B1.** Plain BM25 already ranks a gold record first for 89% (HCMUT) and 78% (labour) of questions, so retrieval cannot separate the variants.
- **B2.** About one third of answerable tasks can be answered without the corpus (closed-book correct for 15 of 45 and 12 of 45; LLM estimate), so we also stratify by closed-book guessability.
- **B3.** 22 HCMUT questions pin a document by a decision number taken from a web listing.

Design and machines. Two benchmarks × three variants × three seeds × 50 tasks give 900 system trials; the two diagnostic baselines below add 190, so the matrix is 1,090 trials, collected as two runs (one per benchmark) of 545 trials each, all recorded. The same model file, `llama.cpp` commit, prompts and budgets are required on both machines; preflight enforces this against `config/benchmark/expected_server.yaml`.

- **Accuracy:** server 82 (shared; one NVIDIA H100 PCIe per run, driver 575.57.08, 16 parallel slots, CUDA build of the same commit; from the run manifests). Latency measured there is not reported.
- **Latency and memory:** Apple M4, 10 cores, 16 GiB, Metal, strictly sequential after warm-up, planned for seed 42 only (300 trials).

Seeds average out the controller's sampling variance (temperature 0.7) but add no tasks, so confidence intervals are taken over tasks. With 50 tasks per benchmark, paired differences below about 10 percentage points (pp) are not detectable.

Diagnostic baselines (not system variants). Two extra cells use the same Bonsai file, sampler, domain prompt, constrained decoding and judge, verification off, seed 42 only, with a disclosed prompt addendum (no tools; only answer, clarify or refuse). *Closed-book:* the question only, no tools and no documents, all 50 tasks per benchmark (100 planned trials); it isolates the parametric-knowledge share of the score. *Oracle:* the full text of the gold records is given up front in the user turn in the read-tool observation format, with no search, on the 45 answerable tasks per benchmark (90 planned trials; refusal tasks have no gold context and are excluded); it isolates the reading ceiling of the 2B controller under perfect retrieval. Both are separate cells, reported only in Tables 10 and 14 beside the system variants restricted to answerable tasks (reference rows), and never pooled with the variants or used in the paired differences (they are not the closed-book probes behind B2).

Error split (answerable tasks). With S a variant's strict success and O the oracle's, $O - S$ is what retrieval, tool use and verification cost relative to gold evidence, and the closed-book rate is the parametric floor. Caveats: the

baselines have one seed and use the addendum prompt, so $O-S$ mixes evidence access with that difference; all values are LLM-judge estimates. The split of failures by whether all gold ids surfaced (`answerable_by_gold_surfaced`) is computed in the analysis summary but not tabulated, so it is not reported.

Metrics. Table 6 defines the metrics.

Metric	Definition (code: `evaluation/metrics.py`, `judge_rules.py`, `stats.py`)
Strict success (LLM-judge estimate)	A trial succeeds only if the run completes, the judgment is valid, behaviour is appropriate, correctness and completeness are both 2/2 and every factual claim is supported (for refusals, absence statements are not claims; a contradicted claim still fails). An execution error counts as failure; an invalid judgment is unknown and also counts as failure. Reported over *all planned* trials, so it is a lower bound, with a 95% bootstrap interval over tasks (seeds averaged per task).
Behaviour confusion (no judge)	Expected behaviour (answer or refuse) against terminal status (answered, refused, clarified, error): over-refusal, unsafe-answer rate, refusal precision and recall, strict match (errors count as mismatches).
Faithfulness	Supported claims over all judged claims (support judged against gold and cited records).
Retrieval (ID proxy)	Recall@5, MRR and nDCG@5 over the agent's trace (distinct ids in order of first appearance, searches then reads), on answerable trials. Plain BM25 on the raw question is near ceiling by B1; the agent's own trace is lower (see Results).
Citation P/R/F1 (ID proxy)	Cited ids against gold ids; nothing cited counts as 0 (macro average). Says nothing about answer correctness.
Errors, latency, tokens	Error rate with Wilson interval by kind; trial and per-call p50/p95 (non-error trials, Mac only); controller tokens from server usage; judge cost as reported by the gateway (not an invoice). Local controller cost is not an API cost, and GPU-active hours are not measured, only wall time and hardware.
Paired difference	Treatment minus baseline per task; paired bootstrap, 5,000 draws, no multiplicity correction.

Table 6: Metric definitions. Strict success, behaviour appropriateness and faithfulness are LLM-judge estimates; retrieval and citation numbers are source-ID proxies, not answer correctness.

Judge and its reliability. DeepSeek V4.1 Flash judges each output through the gateway, blind to the variant and the verifier outcome (thinking on, high effort, JSON validated, up to three attempts plus extra rounds for invalid ones). We claim no seed or temperature, and the model revision behind the route is not pinned [2]. The gateway sees only benchmark questions, candidate answers and gold sources, never user data. LLM judges can be biased [9], so we check this one against human labels. On Saturday 24 October the group labels a stratified blind sample of about 30 items per benchmark with the same claim-counting rule, and `make audit-score` computes agreement and Cohen's κ against the judge's strict-success label. **[PENDING: Human audit not yet performed; agreement and κ unavailable.]** Separating answer correctness from citation support follows Gao et al. [3].

Results (v3, Bonsai 2B). Both benchmarks were collected in full and judged (545 of 545 trials per run: 450 system, 50 closed-book, 45 oracle). HCMUT is run `d2-gpu-main-hcmut-v3` and labour is run `d2-gpu-main-labour-v3` (commit 957579b without tracked changes, harness 1.1.2, one H100 PCIe each, 16 parallel slots). Each table below is rendered from one run by `render_benchmark_tables.py` and names it in its source line; nothing is pooled across benchmarks (in the paired tables the "pooled" rows repeat the benchmark rows, because a run holds one benchmark). Success, behaviour-appropriateness and faithfulness are LLM-judge estimates, not human labels (audit pending); strict success is a lower bound over all planned trials, with 95% intervals over tasks. The Mac latency run has not been done (Table 19 stays red).

Headline. Strict success is low for every variant (Tables 7 and 11). On HCMUT it is 18.0% [11–26] for OFF, 12.0% [7–18] for L1 and 14.0% [8–21] for L1+rerank; on labour it is 14.7% [7–23], 14.7% [7–23] and 18.7% [11–27]. Intervals overlap within each benchmark. Trace Recall@5 is 0.674, 0.659 and 0.756 on HCMUT and 0.607, 0.659 and 0.815 on labour,, below the near-ceiling of BM25 on the raw question (Recall@5 0.99 in an offline probe; B1).

Paired comparisons (Tables 9 and 13; 50 tasks, no multiplicity correction). L1 against OFF changes strict success by −6.0 pp [−13.3, +0.7] on HCMUT and 0.0 pp [−6.7, +6.7] on labour, so no detectable change; the judge-free behaviour match falls by 20.7 pp [−30.7, −10.7] and 22.7 pp [−33.3, −12.0], and the cited-id F1 (a source-ID proxy) by 15.2 and 10.0 pp, with intervals that exclude zero. L1+rerank against L1 raises Recall@5 by 9.6 pp [−0.7, +20.7] on HCMUT and by 15.6 pp [+3.0, +28.9] on labour, but the strict-success differences (+2.0 [−5.3, +9.3] and +4.0 [−5.3, +13.3]) are not significant. So the reranker improves what the agent retrieves on labour (a retrieval proxy) without a measurable gain in judged success.

Answer, refuse, clarify. Of 150 trials per variant, answered/refused/clarified counts are, on HCMUT, 66/25/59 for OFF, 33/59/58 for L1 and 38/55/55 plus 2 errors for L1+rerank; on labour 91/9/49 plus 1 error, 52/45/53 and 49/56/44 plus 1 error (Tables 8 and 12). Over-refusal (refused or clarified when an answer exists) rises from 53.3% for OFF to 77.0% for L1 and 72.2% for L1+rerank on HCMUT, and from 34.3% to 62.2% and 63.4% on labour (Tables 15 and 17). No task expects a clarification, so every clarify on an answerable task is a miss. Unsafe answers on the 15 out-of-scope trials per variant: 20.0%, 13.3%, 6.7% (HCMUT) and 20.0%, 6.7%, 0.0% (labour), from a very small base.

Baselines (Tables 10 and 14; seed 42, not variants). Closed-book strict success is 6.0% on HCMUT (0.0% on the 45 answerable tasks: its three successes are refusal tasks) and 14.0% [6–24] on labour (13.3% [4–24] on answerable

tasks). The oracle, which sees the gold records and needs no retrieval, reaches only 28.9% [16–42] on HCMUT and 44.4% [31–58] on labour, although it answers 84.4% of the answerable tasks: the 2B controller fails the strict criterion on most items even with gold evidence. The oracle is above every variant's answerable-only rate (OFF, L1, L1+rerank: 17.8%, 10.4%, 14.1% on HCMUT; 14.1%, 11.9%, 16.3% on labour), clearly on labour but not clearly on HCMUT (OFF [10–26]). On labour the closed-book interval overlaps every variant's, so there is no evidence that retrieval raised judged success over parametric knowledge.

Execution errors. Five of the 1,090 trials ended in `model_output_error` (invalid or truncated action); there were no timeouts or backend errors. HCMUT L1+rerank had 2 of 150 (1.3%, Wilson 95% [0.4, 4.7]); labour OFF and L1+rerank 1 of 150 each (0.7%, [0.1, 3.7]); labour oracle 1 of 45. They count as failures, never as refusals.

Failure analysis (diagnostics, not measured effects). Computed offline from the recorded ledgers (probes `v3_failure_analysis.py` and `l1_replay.py` in `evaluation/probes`; write-up in `docs/reviews`) with lexical proxies and benchmark reference quotes; associations, not tested interventions, and not for tuning on these benchmarks.

- **Clarify is a missing-evidence exit.** OFF clarifies 50 of 135 answerable HCMUT trials (37%) and 40 of 135 labour trials (30%), and 9 of 15 out-of-scope trials (60%) on both; the oracle clarifies 7 and 6 of 45. On single-evidence tasks OFF clarifies in 17% (HCMUT) and 13% (labour) of trials when the reference evidence was in the model's view, against 58% and 45% when the gold record was not retrieved. A keyword heuristic labels 86 of 90 OFF clarifications on answerable tasks as asking for more document detail and one as asking for a situational fact.
- **Model-written queries retrieve worse than the question.** For the first search of answerable OFF and L1 trials (n = 245 HCMUT, 267 labour), Recall@3 is 0.73 and 0.63 for the model's query against 0.97 and 0.98 for BM25 on the raw question.
- **Repair feedback leads to refusals.** After a verifier repair message the controller refuses in 61 of 109 cases (56%) on HCMUT and 68 of 128 (53%) on labour; only 10 of 90 and 8 of 106 trials with a repair end answered.
- **Quote mismatch dominates L1 rejections.** 172 of the 237 repair events (both benchmarks, L1 and L1+rerank) failed on the exact quote alone. The replay of the unchanged verifier reproduces all 429 recorded drafts. Snippets join clauses with "[...]", which quotes can elide; how much of the mismatch this explains is not isolated.

Reading the results. (i) The L1 variants lower the judge-free behaviour match and raise refusals and clarifications, with no measurable gain in judged success. (ii) Success on closed-book-guessable tasks reflects parametric knowledge as much as grounding (Tables 16 and 18). (iii) Differences under about 10 pp are within noise at 50 tasks, and all success figures are LLM estimates until the audit is done.

Post-hoc revision r2 (in progress). The diagnostics above were read after the v3 results, so any change they motivate is a post-hoc revision evaluated on the same 100 tasks: exploratory, with v3 staying the pre-planned result (`docs/plans/D2_REVISION_L1_BONSAI4B.md`). Revision r2 targets the failure modes above: an L1 verifier v2 that accepts elided and wrapped quotes, checks quantities against the sentence and not the whole record, and returns feedback that shows the closest source span (its support threshold, 0.5, was chosen a priori on the 29 answers of the historical `bll2019` development set, not on these benchmarks, and is a coarse operating point); and controller-side fixes for the first-search query, the clarify and refuse contract, repair framing and observation cues, to be fixed when r2 is frozen. In an offline replay of the recorded drafts, v2 would have accepted 42 of the 237 repaired drafts and would also have rejected 74 of the 172 that v1 accepted (`evaluation/runs/l1-replay-2026-10-24`); lexical proxies, not correctness. r2 will run with Bonsai 2B (isolates the system change) and Bonsai 4B (isolates model size). **[PENDING: r2 results for Bonsai 2B and 4B on both benchmarks (run ids, strict success, paired differences against v3); no r2 run exists yet.]**

Benchmark	Variant	Trials	Succ. (%)	Beh. (%)	Faith. (%)	Recall@5	MRR	Cit. P	Cit. R	Cit. F1
HCMUT-v1	OFF	150	18.0 [11–26]$^{\dagger 8}$	45.1	70.0	0.674	0.554	0.317	0.337	0.317
	L1	150	12.0 [7–18]$^{\dagger 9}$	22.7	64.9	0.659	0.520	0.167	0.167	0.165
	L1+rerank	150	14.0 [8–21]$^{\dagger 11}$	25.2	54.9	0.756	0.630	0.237	0.237	0.235

Table 7: Benchmark results per variant (450 real native trials; seeds 42, 123, 2026; run d2-gpu-main-hcmut-v3). *Succ.* is the strict-success rate (behaviour appropriate, correctness and completeness 2, all claims supported) as a lower bound over all trials, with a task-level bootstrap 95% interval; execution errors count as failures and †k marks k judge-unknown trials counted as failures. Succ., Beh., Faith. are LLM-judge estimates by deepseek/deepseek-v4-flash (gateway-routed, thinking enabled, effort high; model revision not pinned), **not human labels**. The human audit of the judge is **[PENDING]**. Recall@5 and MRR are retrieval of gold ids over the agent's search+read trace (answerable trials; near ceiling on these benchmarks, so they cannot separate variants); Cit. P/R/F1 are macro cited-id set matches to gold ids (nothing cited = 0; source-ID proxies, not answer correctness). Red entries are not measured.

Source: run d2-gpu-main-hcmut-v3 @ 957579b, harness 1.1.2, main.

Cost, latency and reproduction. Judge spend as reported by the gateway, failed attempts included and not an invoice, was USD 0.655 for the HCMUT run and USD 0.628 for the labour run (`analysis/summary.json`). Collection took 946 s and 1,047 s of wall time on the shared H100; GPU-active hours, Mac latency and memory are not measured. Run ids: `d2-gpu-main-hcmut-v3` and `d2-gpu-main-labour-v3`. One command per

Benchmark	Variant	Ans/Ref/Clar/Err	Err. (%)	Trial p50/p95 (s)	Call p50/p95 (s)	tok/s	Tok. in/out	Calls/trial	Unknown	Judge USD
	OFF	66/25/59/0	0.0	*n/r*	*n/r*	*n/r*	5794/472	2.52	8	0.197
HCMUT-v1	L1	33/59/58/0	0.0	*n/r*	*n/r*	*n/r*	6751/593	2.80	9	0.144
	L1+rerank	38/55/55/2	1.3	*n/r*	*n/r*	*n/r*	6351/615	2.65	11	0.158

Table 8: Operational measurements and judge cost per variant. Counts are answered/refused/clarified/execution-error trials; Err. is the execution-error rate. n/r: latency was recorded under concurrent workers and is not reportable. Call = one controller LLM call; tok/s = decode tokens per second from the server timings. Tokens are controller input/output per trial (complete usage only). Judge cost is the gateway-reported usage cost over all retained attempts (all variants: 0.655 USD; not an invoice).

Source: run d2-gpu-main-hcmut-v3 @ 957579b, harness 1.1.2, main.

Comparison	Scope	Tasks	Δ Succ. (pp)	Δ Beh. match (pp)	Δ Rec.@5 (pp)	Δ Cit. F1 (pp)	Δ Err. (pp)
L1 vs OFF	HCMUT-v1	50	-6.0 [-13.3,+0.7]	-20.7 [-30.7,-10.7]	-1.5 [-9.6,+7.0]	-15.2 [-24.2,-6.5]	+0.0 [+0.0,+0.0]
	pooled	50	-6.0 [-13.3,+0.7]	-20.7 [-30.7,-10.7]	-1.5 [-9.6,+7.0]	-15.2 [-24.2,-6.5]	+0.0 [+0.0,+0.0]
L1+rerank vs L1	HCMUT-v1	50	+2.0 [-5.3,+9.3]	+4.0 [-4.7,+12.7]	+9.6 [-0.7,+20.7]	+6.9 [-1.7,+15.8]	+1.3 [+0.0,+3.3]
	pooled	50	+2.0 [-5.3,+9.3]	+4.0 [-4.7,+12.7]	+9.6 [-0.7,+20.7]	+6.9 [-1.7,+15.8]	+1.3 [+0.0,+3.3]
L1+rerank vs OFF	HCMUT-v1	50	-4.0 [-11.3,+3.3]	-16.7 [-26.7,-6.7]	+8.1 [-2.6,+18.5]	-8.3 [-18.1,+1.1]	+1.3 [+0.0,+3.3]
	pooled	50	-4.0 [-11.3,+3.3]	-16.7 [-26.7,-6.7]	+8.1 [-2.6,+18.5]	-8.3 [-18.1,+1.1]	+1.3 [+0.0,+3.3]

Table 9: Paired variant differences (treatment minus baseline, percentage points) with 95% paired bootstrap intervals over tasks (seeds averaged per task; 5000 resamples; pooled = both benchmarks' tasks together). Succ. is the strict-success lower bound (LLM-judge estimate), Beh. the structural behaviour match, Rec.@5 the trace retrieval, Cit. F1 the cited-id F1, Err. the execution-error rate. No multiple-comparison correction; intervals that include 0 show no detectable difference at this sample size.

Source: run d2-gpu-main-hcmut-v3 @ 957579b, harness 1.1.2, main.

Benchmark	Row	Trials	Succ. (all, %)	Succ. (answerable, %)	Answer rate (%)	Cit. P	Cit. R
	closed-book (diagnostic)	50/50	6.0 [0–14]	0.0 [0–0]	11.1	–	–
	oracle (diagnostic)	45/45	28.9 [16–42]	28.9 [16–42]	84.4	0.822	0.811
HCMUT-v1	OFF (reference)	135	–	17.8 [10–26]	46.7	0.317	0.337
	L1 (reference)	135	–	10.4 [5–16]	23.0	0.167	0.167
	L1+rerank (reference)	135	–	14.1 [7–21]	27.4	0.237	0.237

Table 10: Diagnostic baselines, one seed (42), same Bonsai controller, sampler, domain prompt and constrained decoding, *no tools*, verification off; they are not system variants and are never pooled with them. *Closed-book*: question only, all tasks; *oracle*: the full gold records are given up front in the read-tool observation format, answerable tasks only (refuse tasks have no gold context and are excluded). Succ. (all) is over every task the row ran; Succ. (answerable) is restricted to answerable tasks (the like-for-like column) with a task-level bootstrap interval; both are LLM-judge estimates, lower bounds, errors count as failures. Answer rate = answered / non-error trials on answerable tasks. Cit. P/R: cited-id match to gold ids (meaningful for the oracle and the system rows; a closed-book model has no document ids). Reference rows repeat the system variants' answerable-only results (seed-averaged) for comparison; the oracle bounds what a 2B reader achieves with perfect retrieval, closed-book what parametric knowledge alone achieves.

Source: run d2-gpu-main-hcmut-v3 @ 957579b, harness 1.1.2, main.

Benchmark	Variant	Trials	Succ. (%)	Beh. (%)	Faith. (%)	Recall@5	MRR	Cit. P	Cit. R	Cit. F1
	OFF	150	14.7 [7–23]$^{\dagger 2}$	56.1	53.9	0.607	0.547	0.358	0.374	0.358
Labour-VI-v1	L1	150	14.7 [7–23]$^{\dagger 8}$	40.1	62.6	0.659	0.581	0.277	0.256	0.258
	L1+rerank	150	18.7 [11–27]$^{\dagger 12}$	39.1	61.4	0.815	0.685	0.296	0.289	0.289

Table 11: Benchmark results per variant (450 real native trials; seeds 42, 123, 2026; run d2-gpu-main-labour-v3). *Succ.* is the strict-success rate (behaviour appropriate, correctness and completeness 2, all claims supported) as a lower bound over all trials, with a task-level bootstrap 95% interval; execution errors count as failures and $\dagger k$ marks k judge-unknown trials counted as failures. Succ., Beh., Faith. are LLM-judge estimates by deepseek/deepseek-v4-flash (gateway-routed, thinking enabled, effort high; model revision not pinned), **not human labels**. The human audit of the judge is **[PENDING]**. Recall@5 and MRR are retrieval of gold ids over the agent's search+read trace (answerable trials; near ceiling on these benchmarks, so they cannot separate variants); Cit. P/R/F1 are macro cited-id set matches to gold ids (nothing cited = 0; source-ID proxies, not answer correctness). Red entries are not measured.

Source: run d2-gpu-main-labour-v3 @ 957579b, harness 1.1.2, main.

Benchmark	Variant	Ans/Ref/Clar/Err	Err. (%)	Trial p50/p95 (s)	Call p50/p95 (s)	tok/s	Tok. in/out	Calls/trial	Unknown	Judge USD
	OFF	91/9/49/1	0.7	*n/r*	*n/r*	*n/r*	5945/545	2.67	2	0.215
Labour-VI-v1	L1	52/45/53/0	0.0	*n/r*	*n/r*	*n/r*	7539/667	3.13	8	0.142
	L1+rerank	49/56/44/1	0.7	*n/r*	*n/r*	*n/r*	7542/735	3.17	12	0.143

Table 12: Operational measurements and judge cost per variant. Counts are answered/refused/clarified/execution-error trials; Err. is the execution-error rate. n/r: latency was recorded under concurrent workers and is not reportable. Call = one controller LLM call; tok/s = decode tokens per second from the server timings. Tokens are controller input/output per trial (complete usage only). Judge cost is the gateway-reported usage cost over all retained attempts (all variants: 0.628 USD; not an invoice).

Source: run d2-gpu-main-labour-v3 @ 957579b, harness 1.1.2, main.

Comparison	Scope	Tasks	Δ Succ. (pp)	Δ Beh. match (pp)	Δ Rec.@5 (pp)	Δ Cit. F1 (pp)	Δ Err. (pp)
L1 vs OFF	Labour-VI-v1	50	-0.0 [-6.7,+6.7]	-22.7 [-33.3,-12.0]	+5.2 [-0.7,+11.9]	-10.0 [-17.7,-2.6]	-0.7 [-2.0,+0.0]
	pooled	50	-0.0 [-6.7,+6.7]	-22.7 [-33.3,-12.0]	+5.2 [-0.7,+11.9]	-10.0 [-17.7,-2.6]	-0.7 [-2.0,+0.0]
L1+rerank vs L1	Labour-VI-v1	50	+4.0 [-5.3,+13.3]	-2.0 [-12.7,+8.7]	+15.6 [+3.0,+28.9]	+3.1 [-6.5,+13.3]	+0.7 [+0.0,+2.0]
	pooled	50	+4.0 [-5.3,+13.3]	-2.0 [-12.7,+8.7]	+15.6 [+3.0,+28.9]	+3.1 [-6.5,+13.3]	+0.7 [+0.0,+2.0]
L1+rerank vs OFF	Labour-VI-v1	50	+4.0 [-4.0,+12.0]	-24.7 [-36.0,-14.0]	+20.7 [+6.7,+35.2]	-6.9 [-19.9,+5.7]	+0.0 [-2.0,+2.0]
	pooled	50	+4.0 [-4.0,+12.0]	-24.7 [-36.0,-14.0]	+20.7 [+6.7,+35.2]	-6.9 [-19.9,+5.7]	+0.0 [-2.0,+2.0]

Table 13: Paired variant differences (treatment minus baseline, percentage points) with 95% paired bootstrap intervals over tasks (seeds averaged per task; 5000 resamples; pooled = both benchmarks' tasks together). Succ. is the strict-success lower bound (LLM-judge estimate), Beh. the structural behaviour match, Rec.@5 the trace retrieval, Cit. F1 the cited-id F1, Err. the execution-error rate. No multiple-comparison correction; intervals that include 0 show no detectable difference at this sample size.

Source: run d2-gpu-main-labour-v3 @ 957579b, harness 1.1.2, main.

machine runs the whole pipeline: `make benchmark-d2-gpu` on server 82 (see `docs/RUNBOOK_D2.md`) and `make benchmark-d2-mac`. Each writes a run directory under `evaluation/runs/` and refuses to start if the lock or the source has drifted; `make judge RUN=<run>` and `make tables RUN=<run>` re-run the judge and the tables. `make`

Benchmark	Row	Trials	Succ. (all, %)	Succ. (answerable, %)	Answer rate (%)	Cit. P	Cit. R
	closed-book (diagnostic)	50/50	14.0 [6–24]	13.3 [4–24]	48.9	–	–
	oracle (diagnostic)	45/45	44.4 [31–58]	44.4 [31–58]	84.4	0.844	0.811
Labour-VI-v1	OFF (reference)	135	–	14.1 [7–23]	65.2	0.358	0.374
	L1 (reference)	135	–	11.9 [5–20]	37.8	0.277	0.256
	L1+rerank (reference)	135	–	16.3 [9–24]	36.3	0.296	0.289

Table 14: Diagnostic baselines, one seed (42), same Bonsai controller, sampler, domain prompt and constrained decoding, *no tools*, verification off; they are not system variants and are never pooled with them. *Closed-book*: question only, all tasks; *oracle*: the full gold records are given up front in the read-tool observation format, answerable tasks only (refuse tasks have no gold context and are excluded). Succ. (all) is over every task the row ran; Succ. (answerable) is restricted to answerable tasks (the like-for-like column) with a task-level bootstrap interval; both are LLM-judge estimates, lower bounds, errors count as failures. Answer rate = answered / non-error trials on answerable tasks. Cit. P/R: cited-id match to gold ids (meaningful for the oracle and the system rows; a closed-book model has no document ids). Reference rows repeat the system variants' answerable-only results (seed-averaged) for comparison; the oracle bounds what a 2B reader achieves with perfect retrieval, closed-book what parametric knowledge alone achieves.

Source: run d2-gpu-main-labour-v3 @ 957579b, harness 1.1.2, main.

Benchmark	Variant	E=answer: A/R/C/Err	E=refuse: A/R/C/Err	Over-refusal (%)	Unsafe answer (%)	Refusal P (%)	Refusal R (%)	Strict match (%)
	OFF	63/22/50/0	3/3/9/0	53.3	20.0	12.0	20.0	44.0
HCMUT-v1	L1	31/55/49/0	2/4/9/0	77.0	13.3	6.8	26.7	23.3
	L1+rerank	37/51/45/2	1/4/10/0	72.2	6.7	7.3	26.7	27.3

Table 15: Structural behaviour (terminal status vs the frozen expected behaviour, no judge): rows E=answer / E=refuse give counts of answered/refused/clarified/error. Over-refusal = refused or clarified when an answer exists; unsafe answer = answered when the task should be refused; refusal P/R treat *refused* as the positive class (non-error trials); strict match counts errors as mismatches.

Source: run d2-gpu-main-hcmut-v3 @ 957579b, harness 1.1.2, main.

Benchmark	Variant	Closed-book guessable (%)	Not guessable (%)	Refusal tasks (%)
	OFF	17.8 [7–31] (n=15)	17.8 [8–29] (n=30)	20.0 [0–47] (n=5)
HCMUT-v1	L1	8.9 [2–18] (n=15)	11.1 [4–19] (n=30)	26.7 [7–47] (n=5)
	L1+rerank	22.2 [9–38] (n=15)	10.0 [3–18] (n=30)	13.3 [0–40] (n=5)

Table 16: Strict success (LLM-judge estimate, lower bound, errors count as failures) stratified by whether a strong model answered the task correctly *without* the corpus (closed-book probe, LLM estimate, config/benchmark/strata.yaml): guessable tasks measure parametric knowledge as much as grounding. Not guessable = partially correct or unsolved closed-book. Refusal tasks have no closed-book stratum. Intervals are task-level bootstrap; n = tasks per stratum.

Source: run d2-gpu-main-hcmut-v3 @ 957579b, harness 1.1.2, main.

Benchmark	Variant	E=answer: A/R/C/Err	E=refuse: A/R/C/Err	Over-refusal (%)	Unsafe answer (%)	Refusal P (%)	Refusal R (%)	Strict match (%)
	OFF	88/6/40/1	3/3/9/0	34.3	20.0	33.3	20.0	60.7
Labour-VI-v1	L1	51/39/45/0	1/6/8/0	62.2	6.7	13.3	40.0	38.0
	L1+rerank	49/51/34/1	0/5/10/0	63.4	0.0	8.9	33.3	36.0

Table 17: Structural behaviour (terminal status vs the frozen expected behaviour, no judge): rows E=answer / E=refuse give counts of answered/refused/clarified/error. Over-refusal = refused or clarified when an answer exists; unsafe answer = answered when the task should be refused; refusal P/R treat *refused* as the positive class (non-error trials); strict match counts errors as mismatches.

Source: run d2-gpu-main-labour-v3 @ 957579b, harness 1.1.2, main.

Benchmark	Variant	Closed-book guessable (%)	Not guessable (%)	Refusal tasks (%)
	OFF	16.7 [3–31] (n=12)	13.1 [4–23] (n=33)	20.0 [0–47] (n=5)
Labour-VI-v1	L1	5.6 [0–14] (n=12)	14.1 [5–24] (n=33)	40.0 [20–60] (n=5)
	L1+rerank	16.7 [0–36] (n=12)	16.2 [8–25] (n=33)	40.0 [20–60] (n=5)

Table 18: Strict success (LLM-judge estimate, lower bound, errors count as failures) stratified by whether a strong model answered the task correctly *without* the corpus (closed-book probe, LLM estimate, config/benchmark/strata.yaml): guessable tasks measure parametric knowledge as much as grounding. Not guessable = partially correct or unsolved closed-book. Refusal tasks have no closed-book stratum. Intervals are task-level bootstrap; n = tasks per stratum.

Source: run d2-gpu-main-labour-v3 @ 957579b, harness 1.1.2, main.

Benchmark	Variant	Trials	Err. (%)	Trial p50/p95 (s)	Call p50/p95 (s)	Decode tok/s	Rerank call p50 (s)
	OFF	[PENDING]	[PENDING]	[PENDING]	[PENDING]	[PENDING]	–
HCMUT-v1	L1	[PENDING]	[PENDING]	[PENDING]	[PENDING]	[PENDING]	–
	L1+rerank	[PENDING]	[PENDING]	[PENDING]	[PENDING]	[PENDING]	[PENDING]
	OFF	[PENDING]	[PENDING]	[PENDING]	[PENDING]	[PENDING]	–
Labour-VI-v1	L1	[PENDING]	[PENDING]	[PENDING]	[PENDING]	[PENDING]	–
	L1+rerank	[PENDING]	[PENDING]	[PENDING]	[PENDING]	[PENDING]	[PENDING]

Table 19: Skeleton, no Mac run yet: no cell is a result. Latency and memory on the Apple M4 (16 GiB, Metal), run [PENDING]: strictly sequential, warm-up excluded, non-error trials. Trial = whole agent run; call = one controller request; decode tok/s from server timings. The Mac L1+rerank reranker is the GGUF runtime, not the runtime of the accuracy run, so accuracy is never pooled across machines. Peak resident memory over the whole run: llama-server [PENDING] MB, runner [PENDING] MB (sampled).

`benchmark-smoke` output is not reportable.

Provenance of reported tables. Every generated result table carries a machine-readable header and a visible source line with its run id, commit and harness version. Runs are listed in `evaluation/runs/REGISTRY.md` with a freshness status (current, pinned, stale, historical, incomplete); only a report-eligible main run that is current or pinned may feed the report. The two v3 runs are *pinned*: the frozen, pre-planned D2 result at commit 957579b, against which the post-hoc revision r2 will be compared and never merged. `make report-check` refreshes the registry (`make runs`) and refuses tables from a stale, smoke, incomplete or ineligible run, and hand-written numeric tables without a provenance marker; it passed on 24 October 2026 with the 12 generated tables from the two v3 runs. Earlier attempts v1 and v2 of each benchmark on server 82 were superseded by v3; their

directories are not synced to this machine and no number from them is reported. **[PENDING: Re-run `make report-check` on the final tree once r2 is added.]**

6 Threat Model (R6)

Assets. (A1) Faithfulness: answers and citations that users rely on must trace to source sections [7]. (A2) User data: questions, pasted documents, memory, audit log and web-run store. (A3) Integrity of corpora, domain profiles, prompts, model weights and the frozen benchmark and evidence. (A4) The tool boundary: no unreviewed write or egress. (A5) Availability of a bounded local machine. (A6) The judge-gateway credential.

Attacker. The attacker can submit arbitrary questions and pasted documents through the UI or API (instruction override, delimiter spoofing, prompt or memory exfiltration, requests to evade a regulation), including content copied from a hostile third party. The attacker cannot change code, corpora, configs, weights, the ledger or the trusted-UI approval state, and cannot reach loopback servers or other users' sessions. *Non-goals:* compromised host or supply chain, poisoned official sources, multi-tenant isolation, model jailbreaks beyond the listed mitigations, and the correctness of the regulations themselves.

Entry points and mitigations. Table 20 gives the mitigation and residual risk for each entry point.

Entry point (asset)	Mitigation (code)	Residual risk
Question, pasted document (A1, A4)	Pre-model pattern checks, Unicode and delimiter sanitising, untrusted wrapper tags, fixed action space without a write action (`src/security/guardrails.py`, domain profiles)	Paraphrase bypass; Vietnamese and English patterns only; benign false positives; not attack-tested in D2.
Retrieved text (A1)	Tagged `[OBSERVATION_TOOL_RESULT]`; L1 checks ids, provenance and quotes (`lexical_grounding.py`)	Observations are tagged, not filtered: an injected instruction inside a source document would reach the model.
Ingested documents (A3)	Offline, manual ingestion; URL, SHA-256 and pages per record; benchmark locks hash the corpora	No signature check; trust in the official site; image-only files excluded.
Memory (A1, A2)	Disabled in D2 runs; demo reuse off; index version, similarity, re-verification, PII patterns	Earlier-user poisoning; incomplete PII filter.
Config, weights, binaries (A3)	Pinned SHA-256 and commit, preflight identity checks, loopback-only endpoints	Third-party weights and the `llama.cpp` fork are not audited.
Judge gateway egress (A2, A6)	Offline only; benchmark data only; key from environment, never in the repository; blind packets; JSON and quote validation; human audit	Candidate text could steer the judge (untested); route not pinned.
Web UI and API, outbox (A4)	Loopback bind, session cookie and CSRF, size bounds; proposal-only tools; host approval bound to the payload	Single-user assumption; no external sender exists.
Loops, long inputs (A5)	Step, tool, repair, token and wall-time limits; trial watchdog	Errors are failures, not defences.
Web pages	Not applicable: there is no browsing tool.	—

Table 20: Threat model by entry point. Mitigations are implementation boundaries, not measured attack effectiveness; the full-system injection tests of the final report [1, 4] are **[PENDING: planned for W47; no D2 numbers]**.

7 Weekly Plan to the First Final Presentation

Table 21 runs to the first final presentation in week 49.

Week (dates)	Owner (proposed)	Deliverable and acceptance
Sat 24 Oct	All	Human audit of the judge sample; agreement and κ entered in Tables 7 and 11.
Sun 25 Oct, Wed 28 Oct	Group leader, all	Submit report and slides; presentation, live demo, backup video.
W45 (2–8 Nov)	Quyền, Phong	HR walkthrough and findings; harder benchmark slice (unnamed documents, paraphrases) for B1.
W46 (9–15 Nov)	Dũng, Quyền	Failure analysis; verifier false accepts and rejects; reranker parity report.
W47 (16–22 Nov)	Phong, Dũng	Entry-point attack tests and permission and fault checks.
W48 (23–29 Nov)	All	Frozen-artifact reproduction, demo hardening, final report; L2/NLI and vector index only after core evidence.
W49 (30 Nov–2 Dec)	All	Tagged release and Round 1 presentation on 2 December.

Table 21: Proposed plan. **[PENDING: Team to confirm owners, the walkthrough date and the commitments.]**

Riskiest assumptions and fallback. A 2B ternary controller may not answer correctly within its budget, and the benchmarks may be too easy for the variants to differ (B1, B2). If so, we keep the loop, the frozen

benchmarks, the variants and the pipeline, report the measured numbers as they are, and cut the vector index, L2/NLI and further domains.

Self-assessment: at risk. The design is frozen and the pipeline has produced two complete, judged main runs, but their result is weak: strict success of 12.0–18.7% for all variants, with the gold-evidence oracle at 28.9% and 44.4%, and the judge is not yet audited by humans. **[PENDING: Re-assess after the audit and the r2 runs.]**

8 Member Contributions and AI-Use Statement

Member	D1 responsibility (not a completed-work claim)	Evidence of contribution
Lê Trí Quyền	Retrieval and corpus	**[PENDING: team to supply]**
Võ Phạm Tuấn Dũng	Agent, verifier and evaluation	**[PENDING: team to supply]**
Nguyễn Hoàng Phong	Tools, security and UI	**[PENDING: team to supply]**

Table 22: Member contributions. Evidence must match the commit history; none is supplied, and the git history shows a single author account.

AI tools used (Table 23). Their outputs are AI judgments, not human checks. **[PENDING: Team to confirm the list and who used which tool.]**

Tool	Used for
Claude (Claude Code with Sonnet and Haiku sub-agents)	Benchmark drafting, verification, adjudication and QA review; code and report drafting
OpenAI Codex; GitHub Copilot	Earlier code and report revisions; D1 draft
Gemma-4-31B	Oracle and closed-book sanity probes (LLM estimates)
DeepSeek V4 Pro, Gemini 3 Flash	Second opinions on benchmark QA
DeepSeek V4.1 Flash	Evaluation judge

Table 23: AI tools used in the project. The controller and reranker are system components (Table 2).

Verified by the group. **[PENDING: Members to state what they checked themselves (audit labels, source spot-checks, code review, demo rehearsal); nothing is claimed here.]** Automated tests run real implementations, no mocks. **[PENDING: Test count from a recorded run at the frozen commit.]**

References

[1] Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024.

[2] DeepSeek. Your first API call: Model names and legacy alias mapping, 2026. URL https://api-docs.deepseek.com/. Accessed 22 October 2026; documents V4.1-Flash-1017 serving behind the legacy Flash alias.

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